

AI FOUNDATION & MODERN AI LAND SPACE

- AI is a branch of CSE it allow the machine to learn or perform tasks that require to human intelligence such as learning , reasoning , decision-making ,and problem – solving
- AI is used in many fields like social media , banking, education and many other fields

1. Evolution of AI

They are stages in Evolution of AI

i) STAGE 1:

In this stage it is called **RULE-BASED AI** it started Exploding (1950s-1970s) in this the machines followed predefined rules , it is not learn from experience

Example:

IF marks > 40 → Pass

ELSE → Fail

ii)STAGE 2:

In this stage it is called **EXPERT SYSTEM** it is started exploding in (1980s) in this AI is used skillful knowledge , it is used in medical background like diagnosis .In industries, AI helps machines work better, detect faults, and improve production.

iii)STAGE 3:

In this stage they presented **machine learning** in the year (1990s-2010) , in this the machine will learn from data , it will improve performance with experience

iv)STAGE 4:

In this stage they developed **DEEP LEARNING** in the year (2010 onwards) .It is neural network (**neural network** means it is an computer system it is inspired by the human brains it will learn pattern from data **makes predictions or decisions.**)

v)STAGE 5:

In this stage they presented **Generative AI (Present)** , it will generate AI can generate text, images, videos, and code.
Examples include ChatGPT and image generators.

2)AI vs Human Intelligence:

AI	HUMAN INTELLIGENCE
It learns from data	It learns from experience
It process information quickly	It will think creatively
It will work continuously	It need rest
It will good at repetitive task	It will good at adapting to new situations

AI can remember enormous amounts of information accurately.	Human memory is limited and can forget details.
It is superhuman in narrow tasks	Human are sometimes better in narrow tasks

AI is powerful but it can not replace human completely human thinking , creativity, and emotions

3) AI AROUND US :

AI is used in daily life:

- Voice Assistants (Google Assistant, Siri)
- YouTube Recommendations
- Netflix Recommendations
- Face Unlock in Smartphones
- Chat bots
- Google Maps Navigation
- Online Shopping Suggestions
- Self-Driving Cars

AI saves time and improves efficiency, it has become a part of our daily life

- Smartphones
- Search Engines(AI-powered search provides direct answers instead of only showing links.)
- Education
- Healthcare
- Social Media

- Banking and Finance
- Transportation
- Shopping and E-Commerce
- Generative AI(One of the biggest recent developments.)
- Agentic AI (Latest Trend)(Traditional AI only answered questions).

4) Machine Learning vs Deep Learning :

- Both are core branch of AI , it focus on enabling computer to learn from data .
 - Both used to make prediction automate decision-making
 - If we understand difference between both it can help to help to choose right approach of a problem .

MACHINE LEARNING :

It help the computer **system to learn patterns** it help to make predictions or decisions without being explicitly programmed by humans.

Types of ML Algorithm :

- **Supervised Learning:** a type of Machine Learning where the **computer learns using labeled data**.
- **Labeled data:** Labeled data means the correct answer is already given.
- **Example:** Email Spam Detection, student Result Prediction, Disease Detection

Unsupervised Learning: is a type of Machine Learning where the computer **learns** from data **without being given the correct answers or labels**.

- Customer Segmentation, Photo Organization, Market Analysis

Reinforcement Learning: Here the system learns by interacting with an environment and receiving rewards or penalties based on its actions. It is mostly used in gaming and decision making.

APPLICATIONS

Spam Emails, Recommendation Systems, Analytics

Deep Learning:

This is an advanced method in ML. This will use Artificial Neural Networks and it will learn data patterns automatically.

EXAMPLES: Image Recognition, Speech Recognition (in this it will recognise the voice into text and identify) Language Processing

WHERE WE USED: Face Unlock in phones, Google Translate, Voice Assistants, ChatGPT, YouTube Recommendations, Self-Driving Cars

TYPES OF DEEP LEARNING:

It covers

- **Convolutional Neural Networks (CNN):** is a special type of neural network in DL mainly used for **images and videos**. Designed to adaptively learn spatial (position or location of an object) hierarchies of features through convolutional (it divides into small parts and identifies the important features) layers.

- **Recurrent Neural Networks:**

It is the best information for sequential data, it is a deep learning network that processes sequential data by remembering previous information and using it to predict or understand the current input. It helps with speech recognition and

language modeling ." It can not remember long sequence information" .

- **Long Short-Term Memory Networks:**

This is advance type in RNN , this will remember **old information for long time** , it will not forget important information. It resolve the RNN problem.

- vanishing gradient problem(not learning old information in training time) It uses to Speech Recognition, Language Translation, Chatbots, Sentiment Analysis

- **Generative Adversarial Networks:** It has two neural network they are **generator**(creating network ,it generate new data , it create fake in images of anything) and **discriminator**(judge)(it check weather the image is fake or real)

EX: AI Image Generation, Face Generation ,Deepfake Technology , photo Enhancement

- **Transformers:**

It is a advanced neural network architecture, It can **understand the relationship between words in text** . it can also remember important information in long sentence. They are the **backbone of models** like GPT(chat gpt) and BERT(Search engines understanding queries.), used in natural language processing(**NLP**).

- Why Transformers Are Powerful?

Old models are process sentence word by word but in this it can remember entire sentence and also it can remember connection between words, it can train faster way, it give better result

APPLICATION: Object Detection , Natural Language Processing, Autonomous Vehicle

Which One to Choose?

- **ML:** when data is structured (fixed format) and limited, and interpretability (is the ability to understand and explain how an AI model arrives at a particular prediction or decision.) is important.
- **DL:** when working with large unstructured (unfixed format) datasets or complex pattern recognition tasks.

5) Neural Networks Conceptually

Neural Network is a machine learning model inspired by the human brain. It learns from data, finds patterns, and makes decisions automatically. performs tasks such as classification, prediction, and decision-making.

Important Components of Neural Network

- **Neuron:** it the basic processing unit.
- **Connections:** it link neurons together. (like wire)
- **Weights :** it decide the importance of each input.
- **Bias:** it help to adjust the output.
- **Activation Function:** Decides whether a neuron should activate or not. (ReLU (Rectified Linear Unit), Sigmoid ,Tanh(Hyperbolic Tangent)
- **Learning Rule:** it is used to Method used to update weights and improve accuracy.

How Neural Network Learns

It have 3 stages

Input Computation: Data is given to the network.

Output Generation: Based on the current parameters, the network generates an output. Network predicts an answers

Iterative Refinement: The network **refines** its output by adjusting weights and biases, gradually improving its performance on diverse tasks. Network **corrects mistakes and improves.**

Importance of Neural Networks

- Identify complex patterns
 - Learn automatically from data
 - Used in AI technologies(Drive Key Technologies)
 - Improve efficiency
 - Backbone of Artificial Intelligence
- Ex: ChatGPT ,Face Recognition ,Self-driving Cars , Google Translate ,Medical Diagnosis

Layers in Neural Network: They are 3 layers

Input Layer: **Receives** data from outside

Hidden Layer: Performs calculations and learns patterns.(most learning happens here)

Output Layer: Produces final answer.

Working of Neural Network

Forward Propagation:

In this the data will move **in forward** direction like

Input layer ->Hidden layer -> Output Layer

In this steps are involved

- Multiply inputs by weights ,
- Add bias ,
- Apply activation function ,

➤ Produce output.

It have formula

LINEAR TRANSFORMATION

$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$ in this where

w = weights

x = inputs

b = bias

ACTIVATION : it means it will check the result got by the neuron and give final output

Backpropagation:

It identify the neural network mistakes and it will correct the mistakes . learning from the mistakes

Loss Calculation:

In this loss(the mistake done by the network) means it will measure the error in the predication . if loss is more the prediction is mistake , if loss is less the prediction is correct . in this we want minimize the loss.

Gradient Calculation:

It tells that how much weight we want to change , it also guide the direction(network) were to change the weight .

Weight Update:

It will improve the network knowledge . it will change the weights based on gradient information. It use optimization algorithm like stochastic gradient descent (SGD) to change.

ITERATION:

It repeating the Forward Propagation + Backpropagation process Again and again .we want need of many iteration because then it will give correct answer . mostly we use in this EMAIL scam

- In features vector it converts words into numbers

Types of Learning

- ❖ **Supervised Learning:** In this Data contains correct answers.

Ex:

Teacher gives: Question and Correct Answer

Network learns from both.

- ❖ **Unsupervised Learning:**

Only input data is given .No answers provided.

EX: voice, images..

- ❖ **Reinforcement Learning:**

It learns form based on it's environment . Learns using rewards and punishments.it approach is mostly used in gaming and decision -making

EX:

Game-playing AI.

Good move → Reward

Bad move → Penalty

Types of Neural Networks

- ❖ **Feedforward Neural Network:**

In this data moves only in forward direction.

Input → Hidden → Output .Simple neural network.

- ❖ **Single Layer Perceptron**

It contains only one / single layer.

- ❖ **Multi-Layer Perceptron (MLP):**

Has multiple hidden layers. Can solve complex problems. It is a type of feedforward neural network with three or more layers, including an input layer, one or more hidden layers and an output layer.

- ❖ **Convolutional Neural Network (CNN):**

It is designed for image processing. It uses convolutional layers to automatically learn features from input images, enabling effective image recognition and classification.

❖ **Recurrent Neural Network (RNN):**

- Handles **sequential data** using feedback loops to retain context over time. Text , speech , language translation

❖ **Long Short-Term Memory (LSTM):**

A type of RNN with memory cells and gates to handle long-term dependencies and avoid vanishing gradients.

Implementation of Neural Network using TensorFlow

Tensor flow : TensorFlow (it look toolbox for ai developer)is an open-source deep learning and machine learning framework developed by [Google](#).

- It helps to build neural network, Deep Learning Models, AI Applications . tensor flow provide ready – made function and libraries. It handles most task automatically.

THEY ARE 6 STEPS

- **Import Necessary Libraries:**

Here are the ready- made tools into our program.

NumPy → handles numbers

Pandas → handles data

TensorFlow/Keras → builds neural networks

NOTE:

“**Keras** is a high-level deep learning library used to build neural networks easily. makes TensorFlow easier to use. Without Keras, writing neural network code is long and complicated. With Keras, you can create a neural network in just a few lines of code. Keras is a high-level API of TensorFlow used to easily create, train, and manage neural network and deep learning models.”

- **Create and Load Dataset:**

Dataset = Collection of examples used for learning.

- **Create a Neural Network:**

We are building the neural network.

Input



Hidden Layer (8 Neurons)



Output Layer (1 Neuron)

- **Compile the Model**
- **Train the Model**
- **Make Predictions**

Advantages

1. Learns automatically from data.
2. Excellent at pattern recognition.
3. Handles complex problems effectively.
4. Supports parallel processing for faster computation.
5. Models complex non-linear relationships in data.

LIMITATIONS

1. High Computational Cost

- Training large neural networks requires powerful computers and GPUs.
- It takes more time and resources.

2. Black Box Nature

- It is difficult to understand how the network makes decisions.
- This can be risky in healthcare, banking, and other critical applications.

3. Overfitting Problem

- The model may memorize training data instead of learning patterns.
- As a result, performance on new data becomes poor.

4. Requires Large Datasets

- Neural networks need a large amount of labeled data for effective training.
- Small datasets often lead to poor performance.

5. Data Quality Matters

- If the training data is biased or incorrect, the predictions will also be inaccurate.

Applications

1. Image and Video Recognition
2. Natural Language Processing (NLP)
3. Finance(Predicting stock prices)
4. Healthcare(diagnosing diseases)
5. Gaming and Autonomous System

6) TRAINING VS INFERENCE

TRAINING:

It is the process where **an AI model learns from data**. It uses large amount of data, model learns patterns.

During training:

- AI is shown many examples.
- It learns patterns.
- It improves its accuracy.

Ex: a student studying book(it is learning stage)

INFERENCE:

It is the process where a trained **AI uses what it has learned** to make predictions or decisions. It uses new data, models use learned patterns .

Ex: A new image is given to the trained AI.

AI predicts: "This is a Cat."

This prediction process is called **Inference**.

AI LEARNING METHODS :

- **Supervised Learning:** Learning with correct answers.
Example: Spam email detection
- **Unsupervised Learning:** it help to find hidden patterns
Example: Grouping similar customers.
- **Reinforcement Learning:** Learning through rewards and penalties.
Example: AI learning to play chess.
- **Self-Supervised Learning:** Learning by predicting missing information.
Example: Language models like ChatGPT predicting the next word.

AI TYPES BY INTERACTION STYLE

- **Conversational AI:** “talk with users”
Conversational AI can talk with people and answer questions in a natural way. It understands what users say and responds like a human conversation.
Example: ChatGPT , customer support chatbots, virtual
- **Agentic AI:** “Acts and completes tasks independently”
Agentic AI can perform tasks, make simple decisions, and complete work with less human help. It can follow a goal and take multiple steps to achieve it.
Example: An AI assistant that schedules meetings and sends reminders automatically.
- **Generative AI:** “creates new content”
Generative AI can create new content such as text, images, videos, music, and code. It generates content based on the instructions given by the user.
Example: ChatGPT, Gemini
- **Autonomous AI:**
Autonomous AI is AI that can perform tasks and make decisions on its own **with little or no human help.**

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It can observe a situation, decide what to do, and take action to achieve a goal.

Ex: self-driving, robot